



VUV/Fe(II)/H₂O₂ as a novel integrated process for advanced oxidation of methyl *tert*-butyl ether (MTBE) in water at neutral pH: Process intensification and mechanistic aspects

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ARTICLE INFO

Article history:

Received 14 June 2019

Received in revised form

13 August 2019

Accepted 5 September 2019

Available online 6 September 2019

Keywords:

Advanced oxidation process

Vacuum ultraviolet light

Fenton reaction

Water contamination

Degradation pathway

ABSTRACT

Vacuum UV (VUV) technologies have recently attracted high interest due to their high efficacy in generating reactive oxygen species (ROS). To date, no systematic study of the modes of action of the integrated VUV/Fe(II)/H₂O₂ process against contaminants elimination exists; the present study reports the oxidation of MTBE in a new light-assisted Fenton-process, by employing either narrowband UVC (254 nm) or VUV (185 and 254 nm) irradiation, in a comparative evaluation. The processes under investigation were the UVC- or VUV/Fe(II)/H₂O₂ sensitized ones and their constituents, i.e. Fe(II)/H₂O₂, VUV, VUV/Fe(II), VUV/H₂O₂, VUV/Fe(II)/H₂O₂, as well as the UVC, UVC/H₂O₂ and UVC/Fe(II)/H₂O₂. We scrutinize the operational parameters of the VUV-assisted process, its enhancements and synergies, comparison with the UVC-based ones, as well as their inflicted pathways towards MTBE degradation. Complete degradation and 87.8% mineralization of 50 mg/L MTBE was achieved in the VUV/Fe(II)/H₂O₂ process (0.9 mM Fe(II) and 3 mM H₂O₂), at near-neutral pH (reaction times: ~30 and 60 min, respectively). Irradiation with VUV light was found to act synergistically and in high kinetic rates enhancement compared to the UVC source, sensitizing the Fenton process for effective oxidation of MTBE in the aqueous solution. A scavenger study and degradation by-products investigation has been performed, leading to a mechanistic pathway proposition, elucidating MTBE degradation. The VUV/Fe(II)/H₂O₂ process demonstrated potential applicability in the field since it could efficiently treat (100% degradation and 86.4% mineralization) groundwater spiked with MTBE, operated either under batch or continuous-flow mode. The findings clearly indicates the VUV-assisted Fenton as an emerging and viable technology for field application to treat the MTBE-contaminated effluents or waters.

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1. Introduction

Methyl *tert*-butyl ether (MTBE, C₅H₁₂O) is an oxygenated additive of gasoline with an annual production rate of 20 Mt around the world (Levchuk et al., 2014). It is released into the environment mainly through leakage/spill/accident during the production, storage and transportation, and use of MTBE and gasoline (Moussavi et al., 2009). Since MTBE has relatively high solubility and a low octanol–water partition coefficient (O'Neil, 2013) it does not adsorb to soil particles thus reaches to the groundwater upon release and is very soluble in water (Pal et al., 2018; Russo et al.,

2015). Since MTBE is resistance to chemical and microbial degradation and consequently low natural attenuation, it is persistent in groundwater and causes the water contamination (Facetti et al., 2019). Accordingly, MTBE is known one of the most important contaminants frequently detected in range of microgram to milligrams per liter in groundwater around the world possessing an increasingly concern (Facetti et al., 2019; Gonzalez-Olmosa et al., 2009; Minetti et al., 2017). MTBE has acute and chronic toxicity and imparts adverse health effects on human be exposed to MTBE through the ingestion, inhalation and skin paths (Roberts et al., 2014; Zhang et al., 2016). In addition, MTBE imparts a very objectionable taste and odor to water (WHO, 2017). EPA advised the taste and odor thresholds of 40 and 20 µg/L, respectively, for MTBE in drinking water (USEPA, 2018).

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